

Evolutionary Robotics

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Objectives of this lecture

Students ...

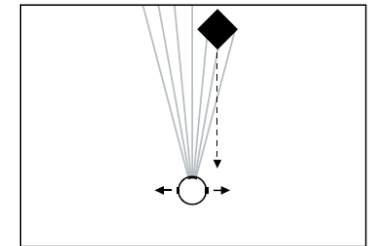
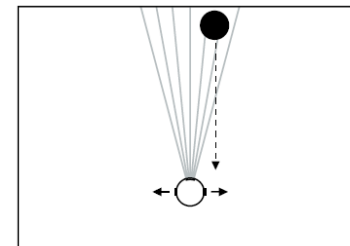
1. Can describe the major components of an evolutionary algorithm (EA), and reason about when it is advisable to apply an EA to a problem.
2. Can explain the major defining properties of Evolutionary Robotics and the related view on cognition, and illustrate this with concrete examples.
3. Can apply an EA to evolve a robot controller and subsequently analyze the found solution.
4. Can explain the difference between reactive and pro-active controllers, and describe limitations of the former.

Recommended literature

Nolfi, Stefano. Power and the limits of reactive agents. *Neurocomputing* 42, 1 (2002): 119-145.



Beer, Randall D. The dynamics of active categorical perception in an evolved model agent. *Adaptive Behavior* 11.4 (2003): 209-243



Evolutionary Algorithms

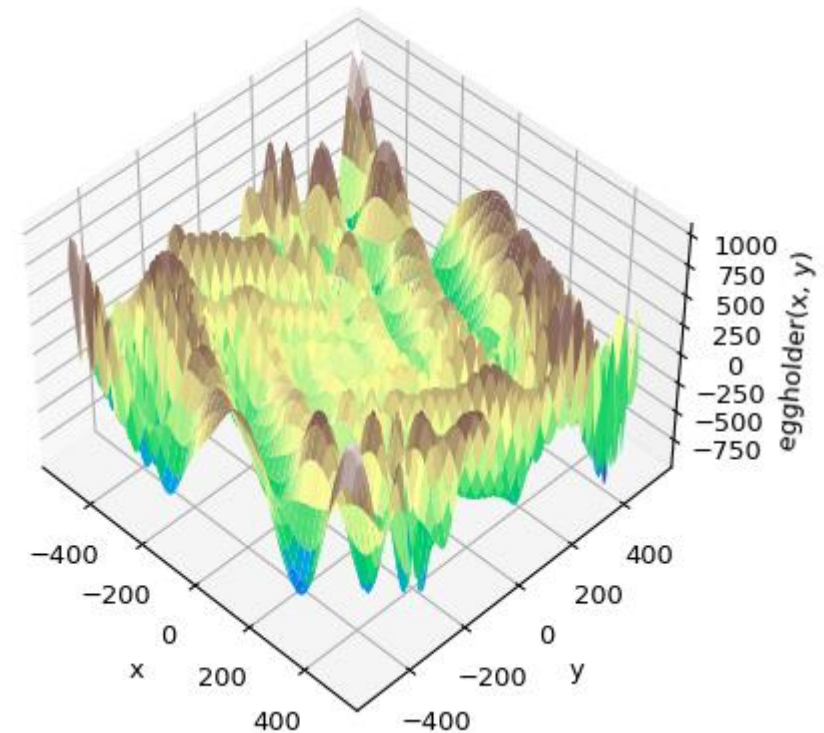
Evolutionary algorithms

are **meta-heuristic optimization** algorithms.

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For example, an EA could find the coordinates (x,y) that maximize the value of the egg-holder function.

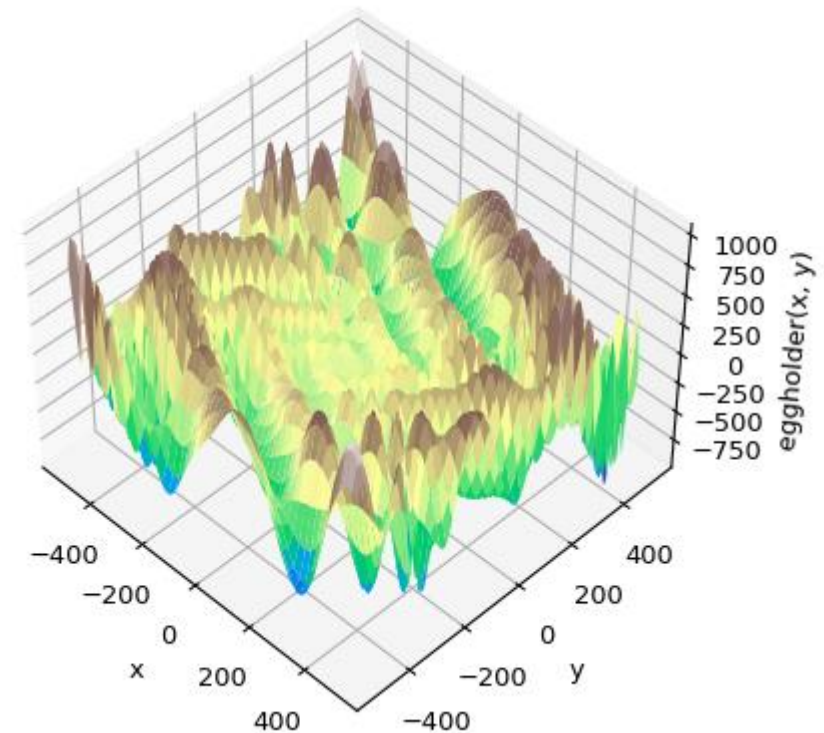


Evolutionary algorithms

are **meta-heuristic optimization** algorithms.

For example, an EA could find the coordinates (x,y) that maximize the value of the egg-holder function.

(X,Y) = the search space



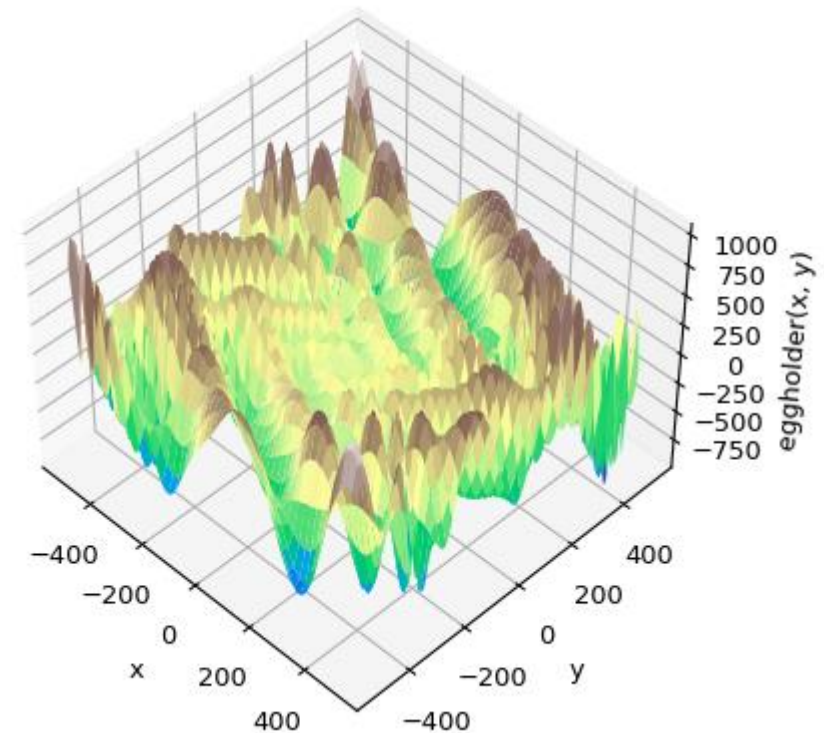
Evolutionary algorithms

are **meta-heuristic optimization** algorithms.

For example, an EA could find the coordinates (x,y) that maximize the value of the **egg-holder function**.

(X,Y) = the search space

Egg-holder = the fitness function



Metaheuristic optimization methods

Metaheuristic optimization methods:

- simulated annealing,
- evolutionary algorithms,
- ant colony optimization,
- particle swarm optimization
- ...

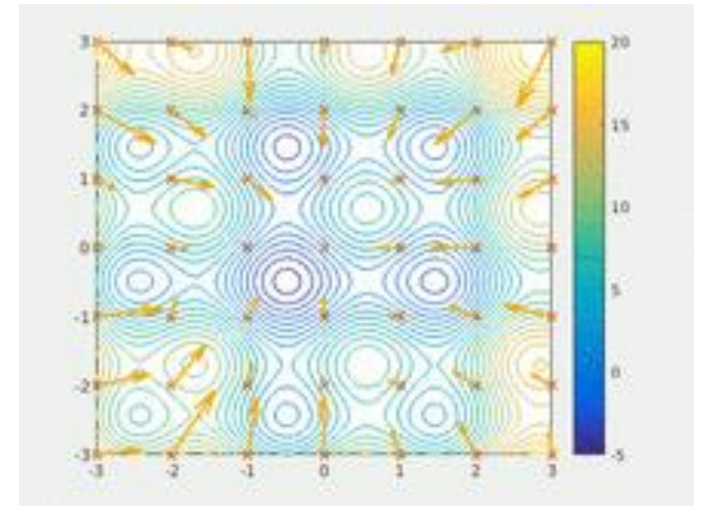
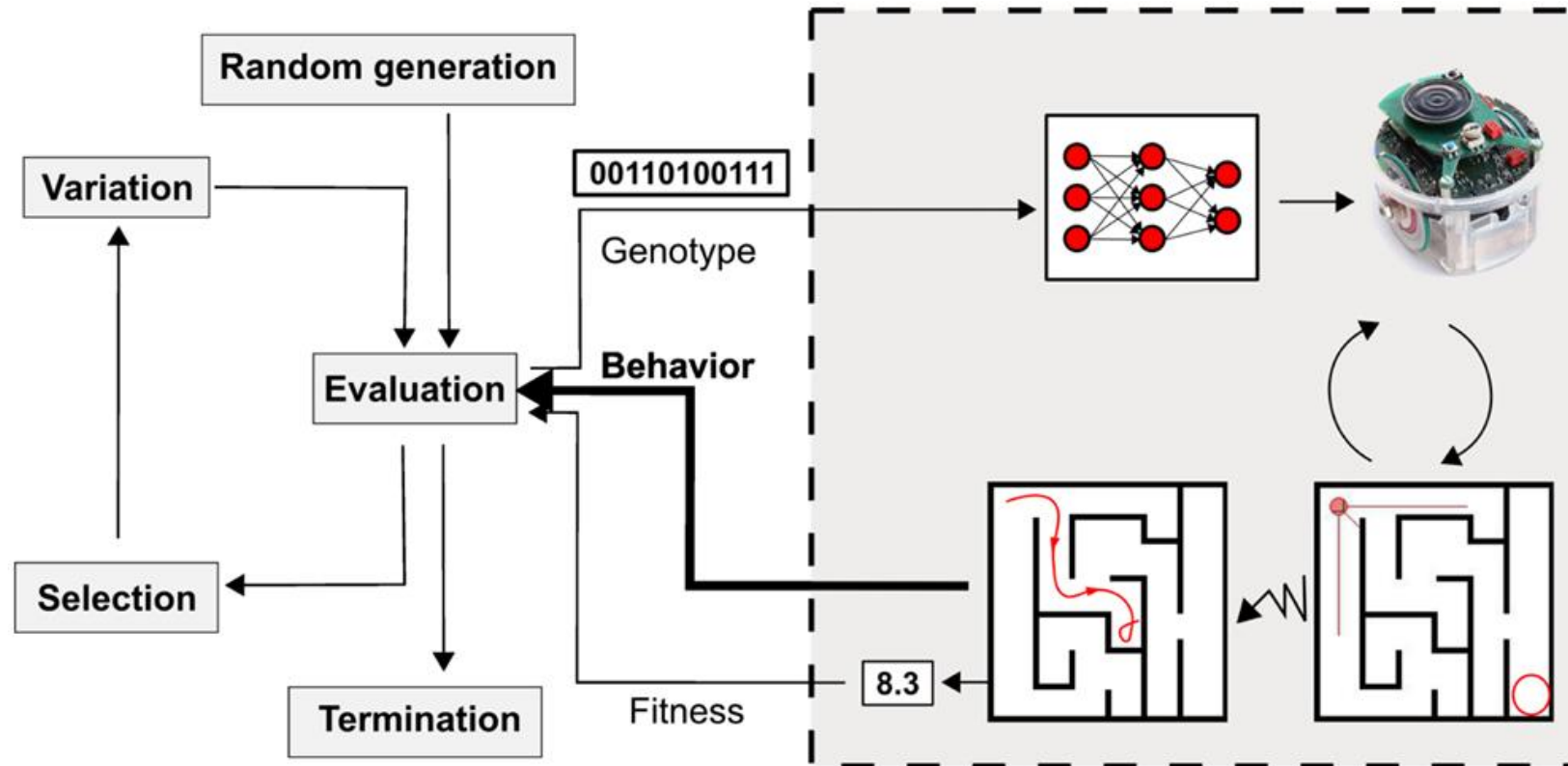


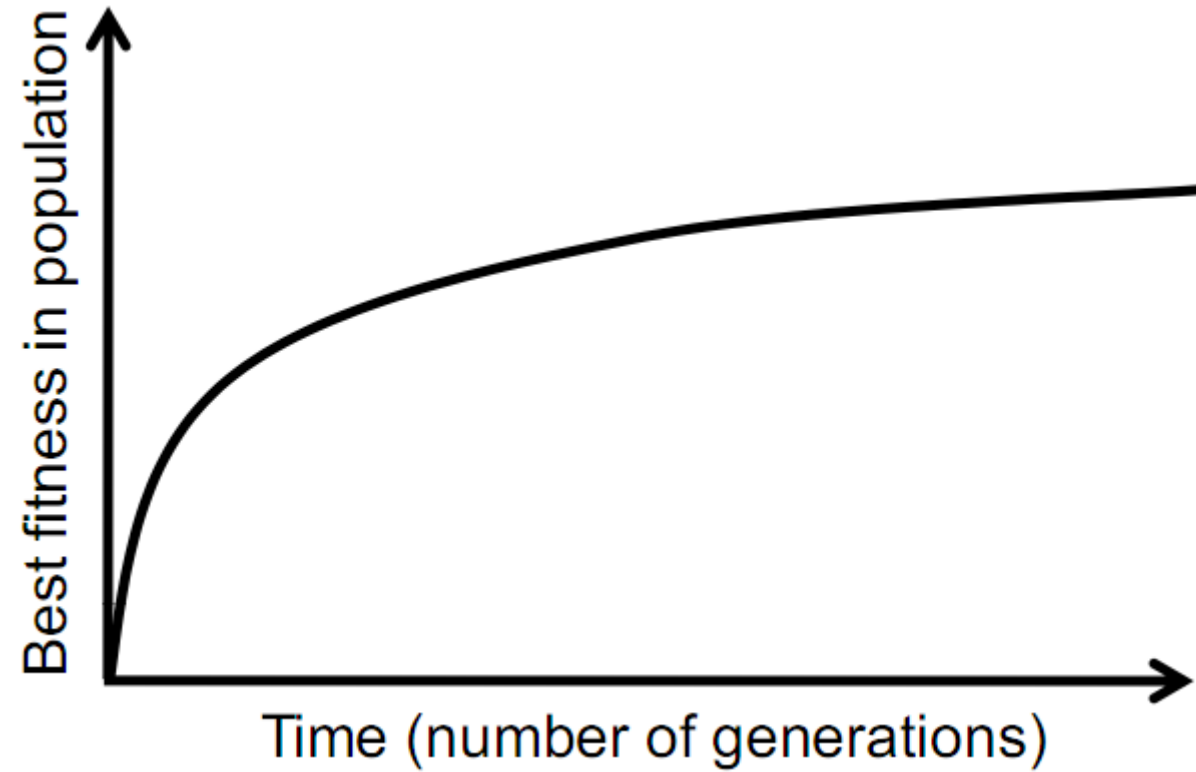
Illustration of PSO on Wikipedia

Typical EA

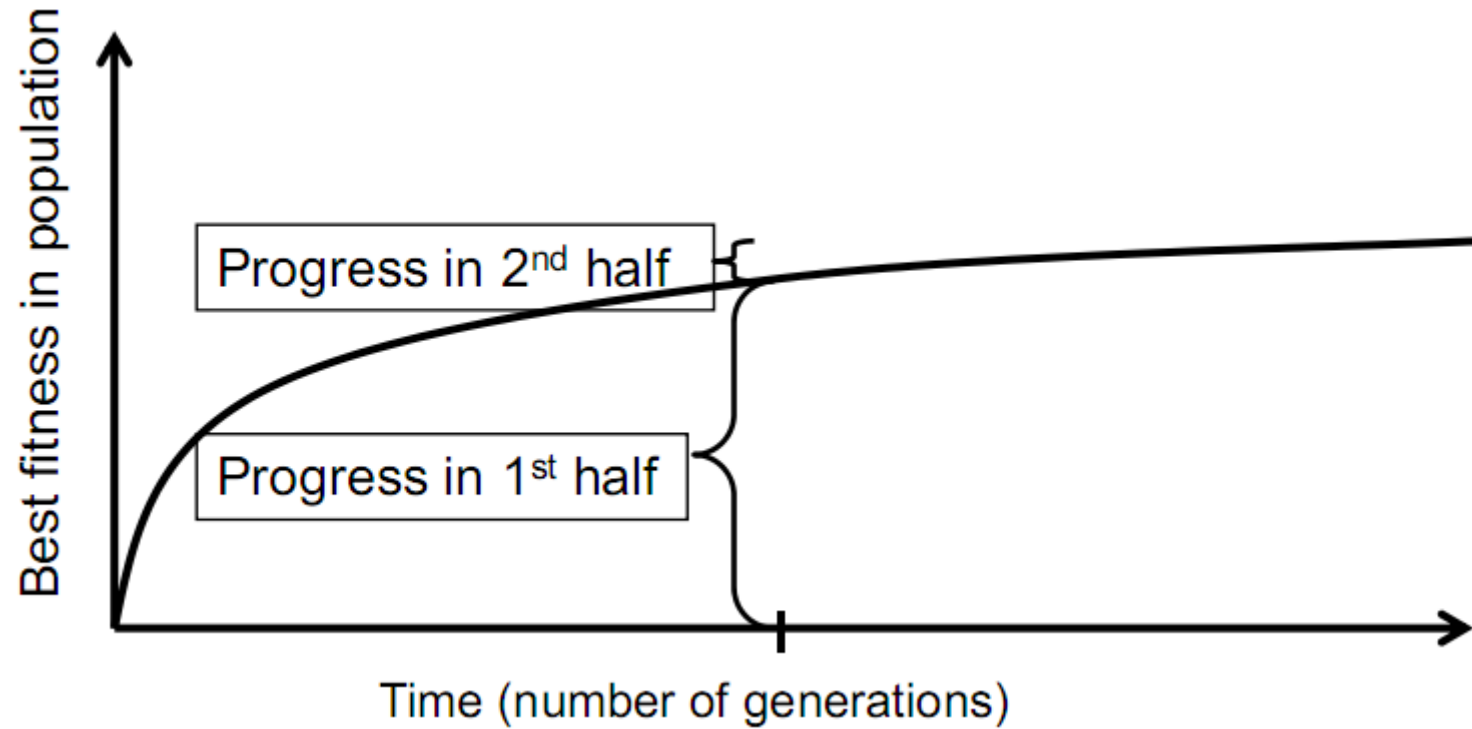


Doncieux, S., Bredeche, N., Mouret, J. B., & Eiben, A. E. G. (2015). **Evolutionary robotics: what, why, and where to.** Frontiers in Robotics and AI, 2, 4.

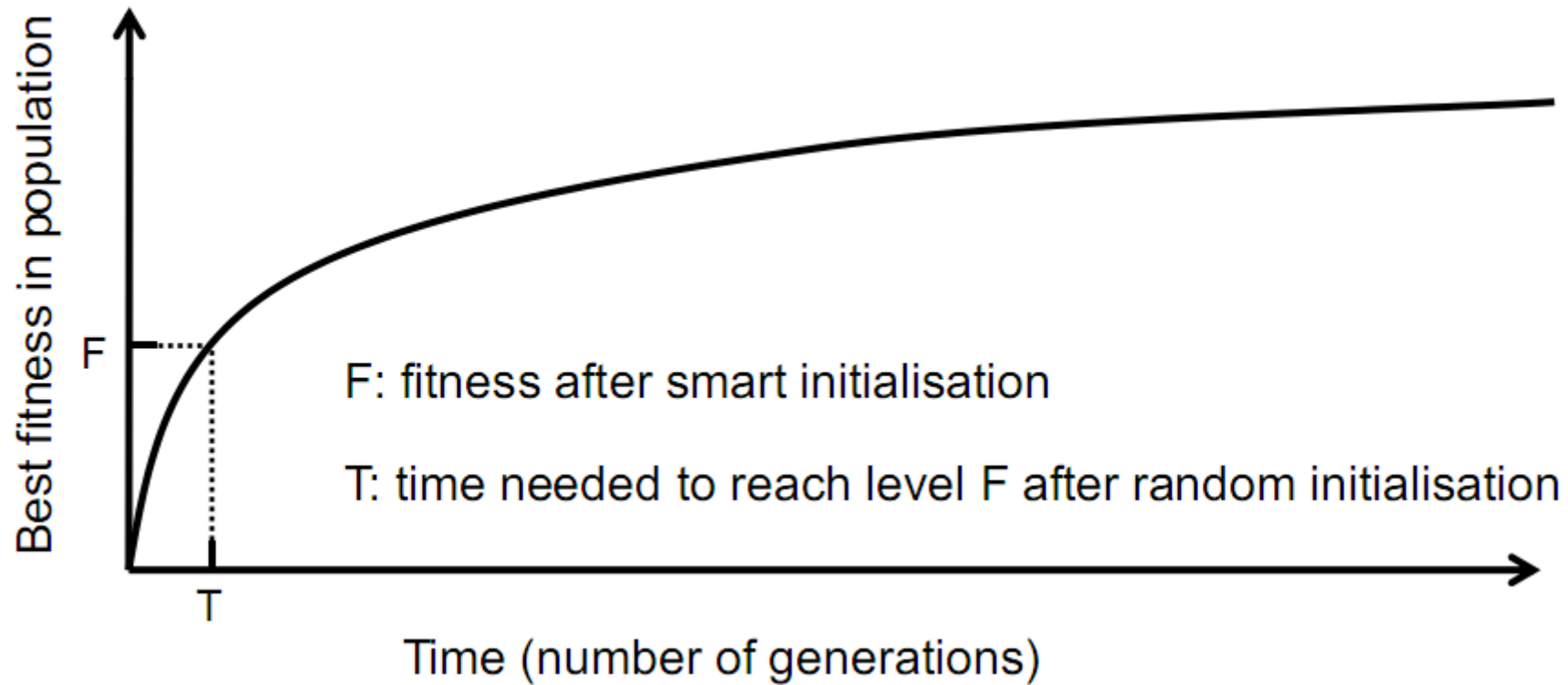
Typical run of an EA



Are long runs beneficial?



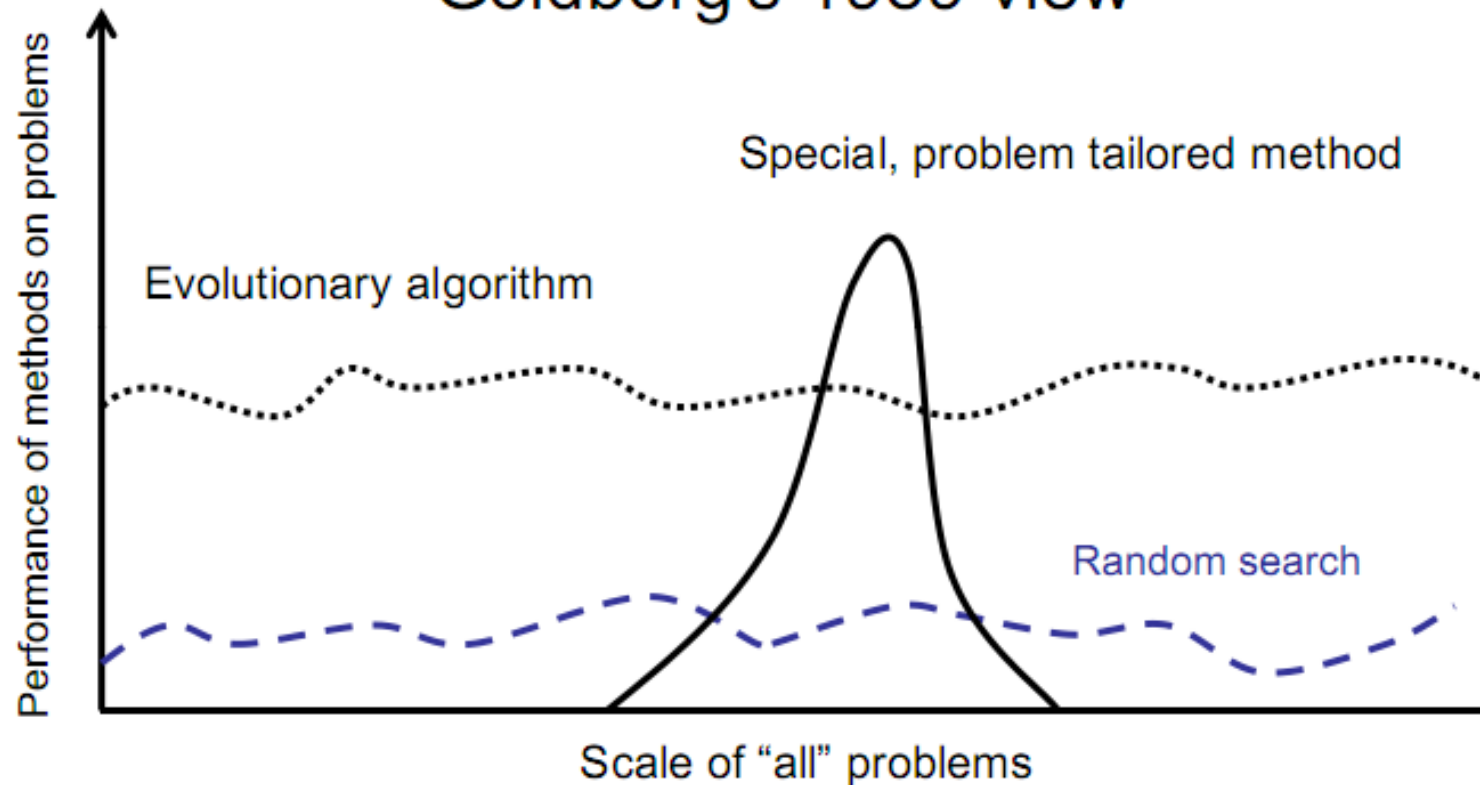
Is smart initialization beneficial?



EAs as problem solvers

Do you agree? Why?

Goldberg's 1989 view



EAs as problem solvers

Goldberg's 1989 view

problems ↑

Special, problem tailored method

No free lunch theorem by Wolpert and Macready

$$\sum_f P(h_m^y | f, m, a_1) = \sum_f P(h_m^y | f, m, a_2)$$

Perform

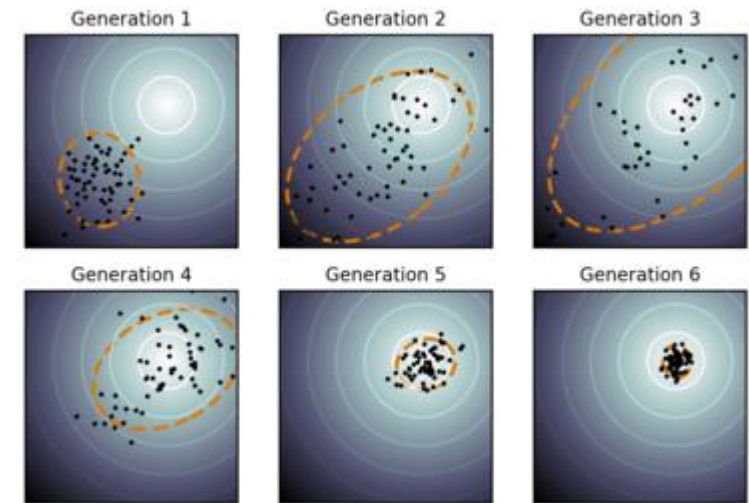
Scale of "all" problems



Example algorithm: CMA-ES

Covariance Matrix Adaptation Evolution Strategy:

- Population is sampled (as in Evolution Strategies) according to a mean and covariance matrix.
- The *mean* is adapted to the successful candidates (*points* in the search space).
- The *covariance matrix* is adapted to maintain the successful search *steps*.



<https://en.wikipedia.org/wiki/CMA-ES>

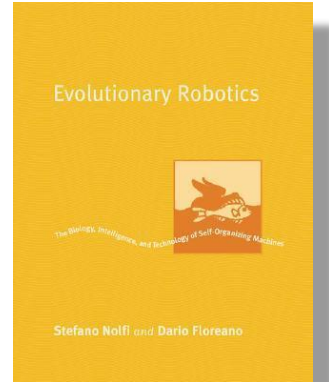
Exercise

Consider a classification task in which we have to learn a function f , mapping a continuous input vector $\vec{x}$ to a target output t . Learning can be based on a training set A , consisting of n pairs $(\vec{x}_i, t_i)$, while testing will happen on a separate set B .

- How would you represent the function?
- What method would you use to optimize the function?
- Would you consider an EA? Why?

Evolutionary robotics

Two main goals of ER

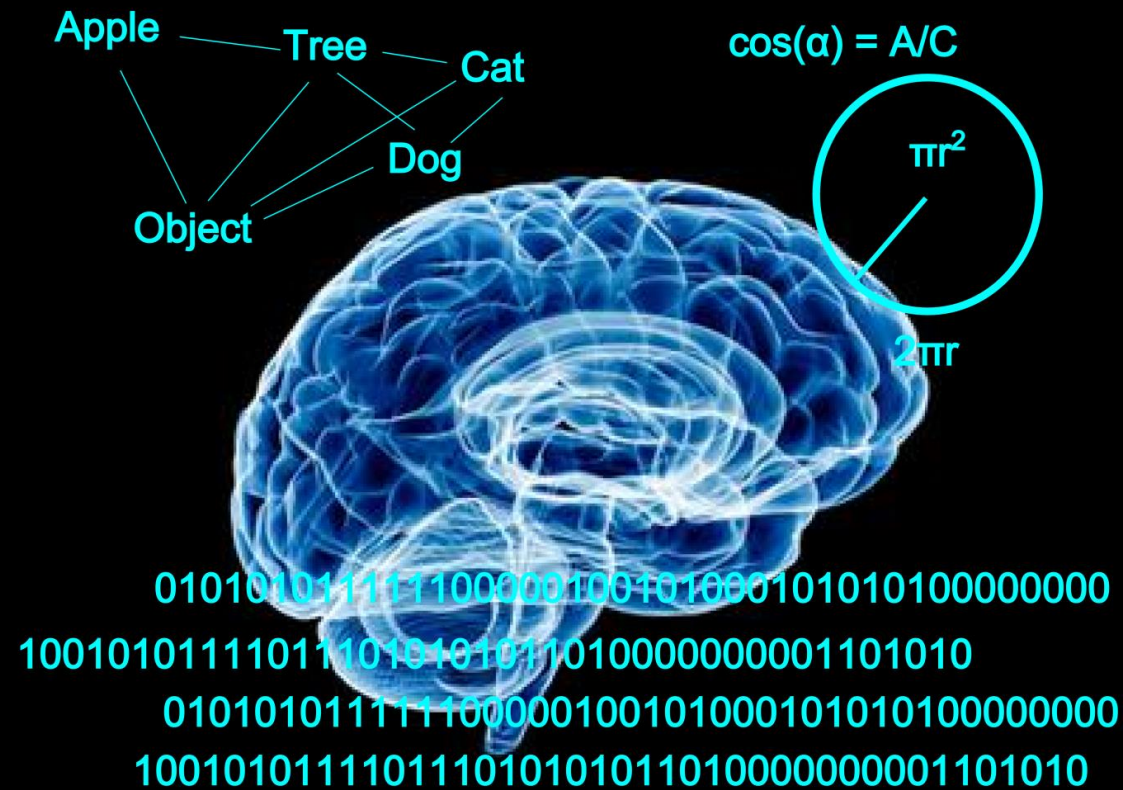


- (1) Understanding generic principles of (embodied) intelligence
- (2) Creating a generic methodology to tackle different tasks

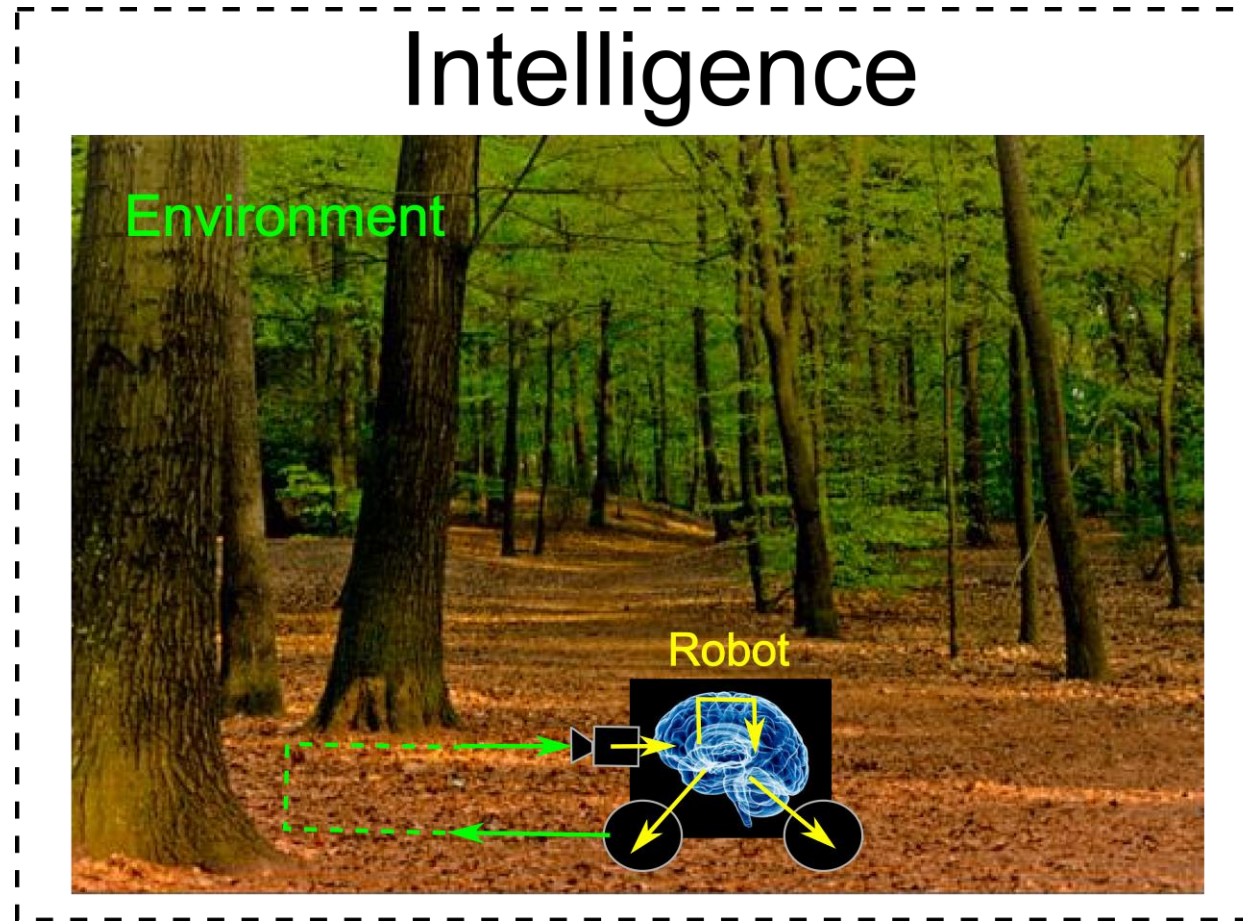
Main defining properties

- Embodied cognition (sensory-motor coordination)
- Self-organization of complex systems
- Metaheuristic optimization of robot controller (*direct policy search* in RL), sensors, actuators, body, ...

Traditional view on cognition

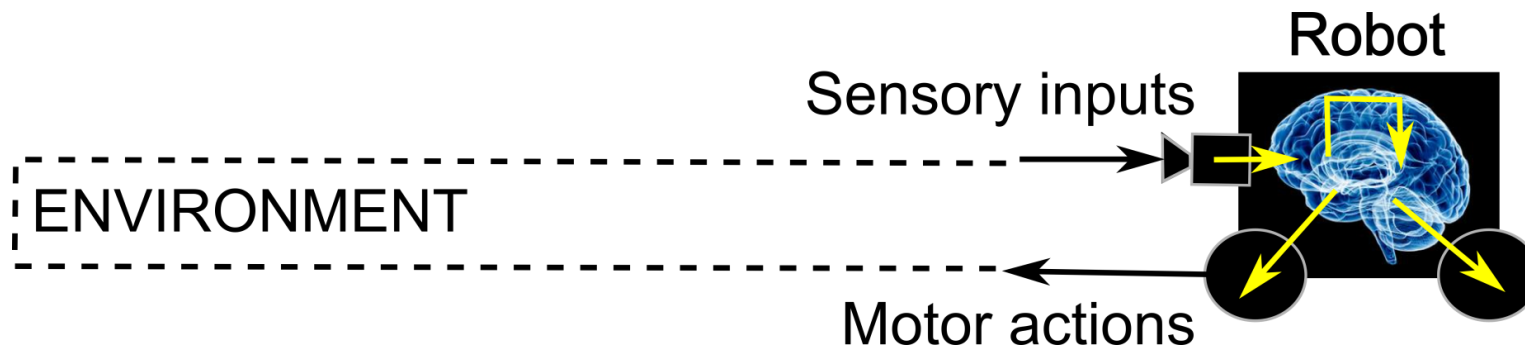


Embodied cognition



Sensory-motor coordination (SMC)

Exploit the feedback loop between sensory inputs and motor actions.



Situated agents

Situated agents:

- Embedded in an environment
- Perceive their environment with their sensors
- Act in their environment with their actuators

→ Closed loop of actions and sensory inputs

Reactive Controllers

Reactive controllers

Reactive controllers:

Controllers without internal state.

Actions a are a function of the sensory inputs $\vec{i}$:

$$\vec{a} = f(\vec{i})$$



Picture from
Caring pets

Perceptual ambiguity



Perceptual Ambiguity

Markov decision problem (MDP):

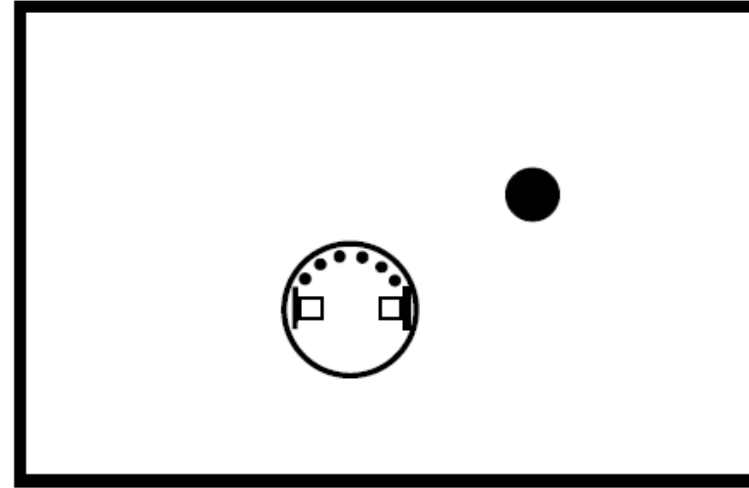
The next state only depends on the current state and action

$$\begin{aligned}\vec{x}_{t+1} &= f(\vec{x}_t, \vec{a}_t) \\ P(\vec{x}_{t+1} | \vec{x}_{1:t}, \vec{a}_{1:t}) &= P(\vec{x}_{t+1} | \vec{x}_t, \vec{a}_t)\end{aligned}$$

Partially observable Markov decision problem (POMDP):

The next state does not only depend on the current state and action

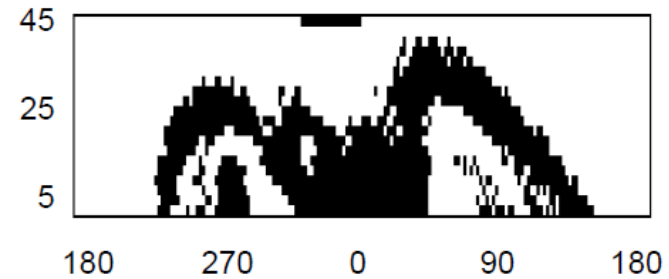
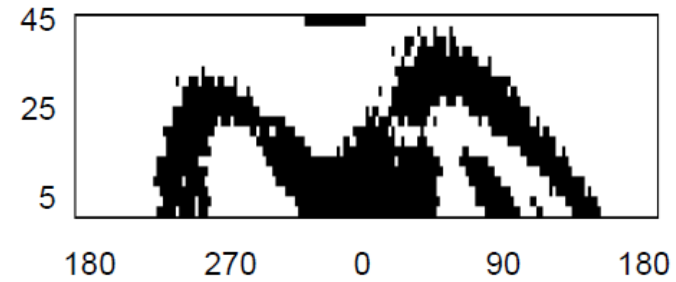
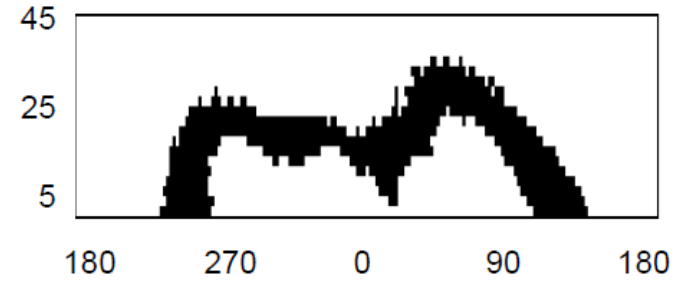
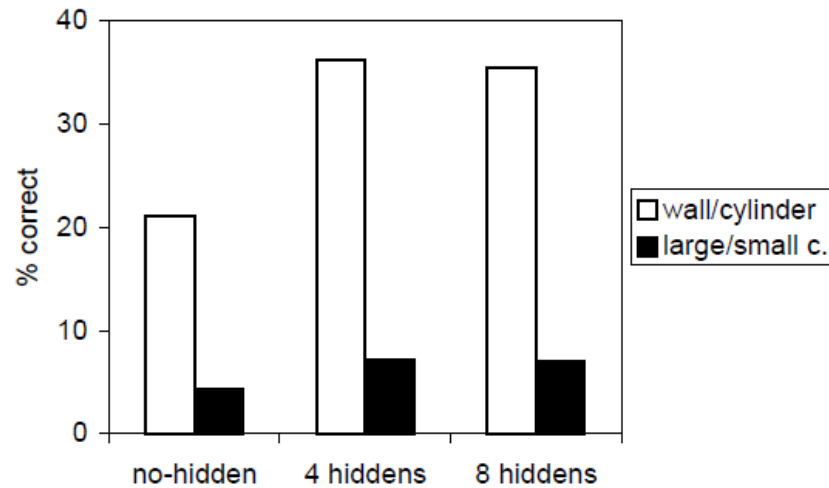
Classification task with Kephera



Kephera has to use its 6 frontal infrared sensors to approach cylinders and avoid walls.

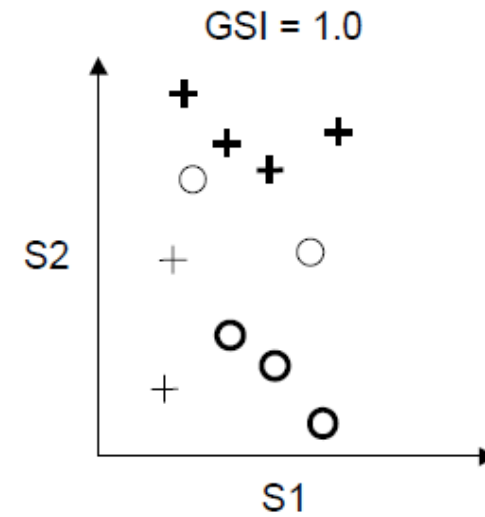
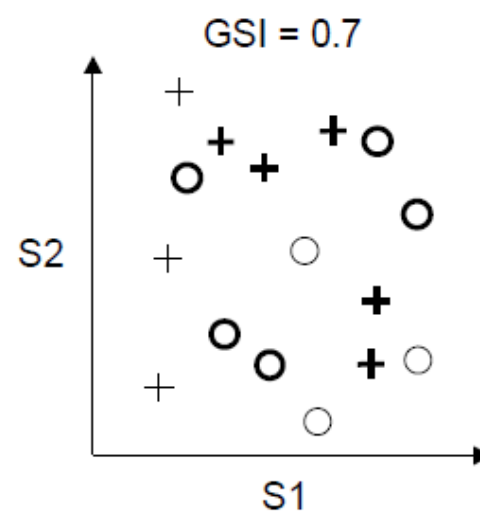
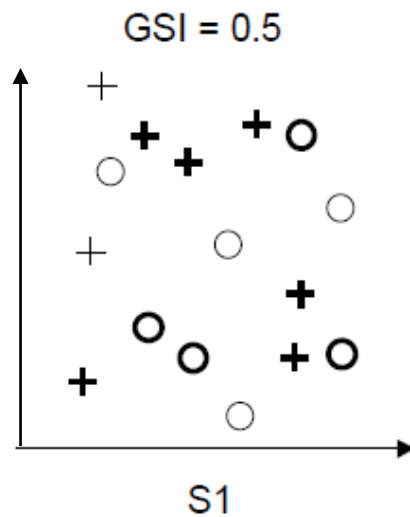
Nolfi, Stefano. "**Power and the limits of reactive agents.**" *Neurocomputing* 42.1 (2002): 119-145.

Passive classification

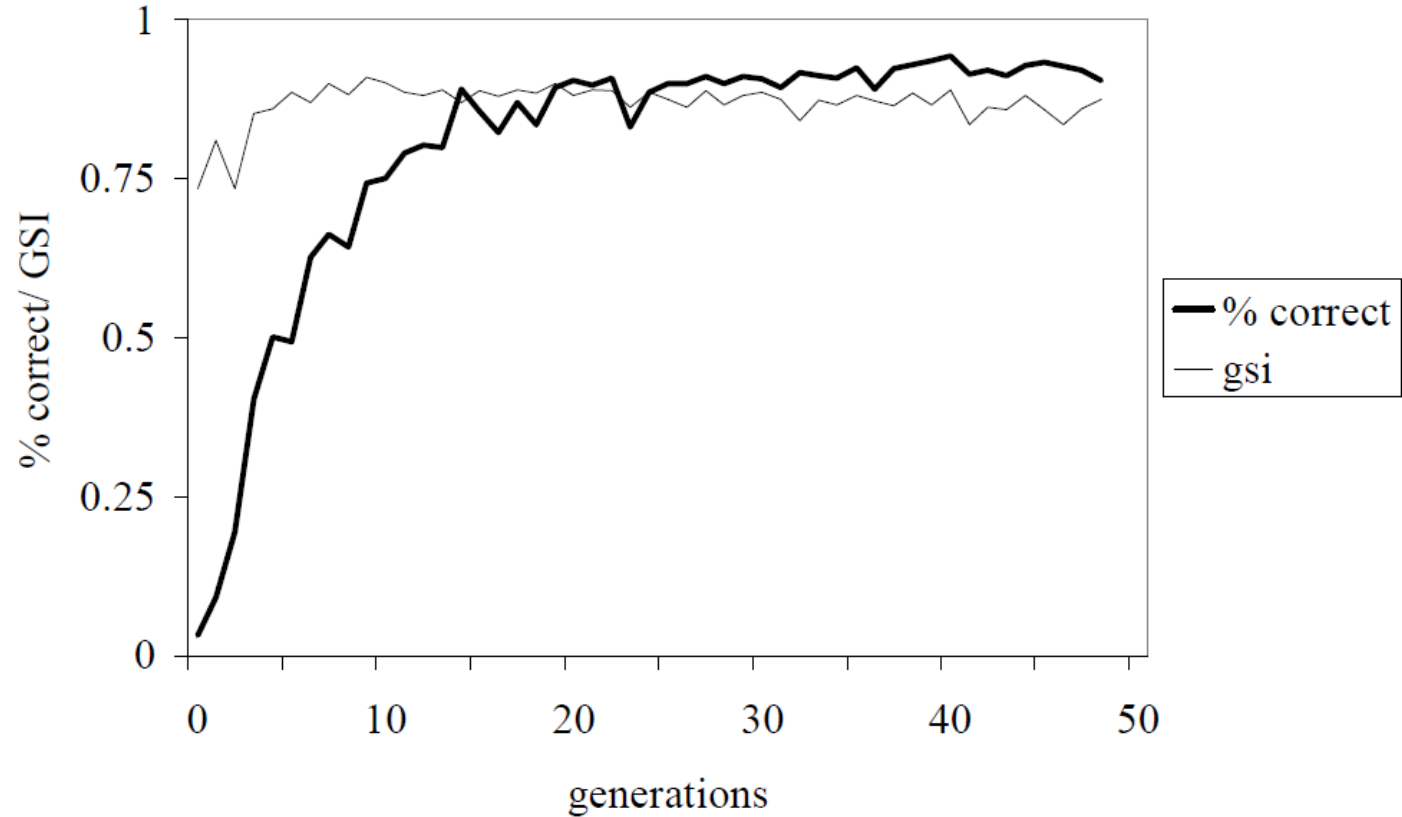


Geometric Separability Index (GSI)

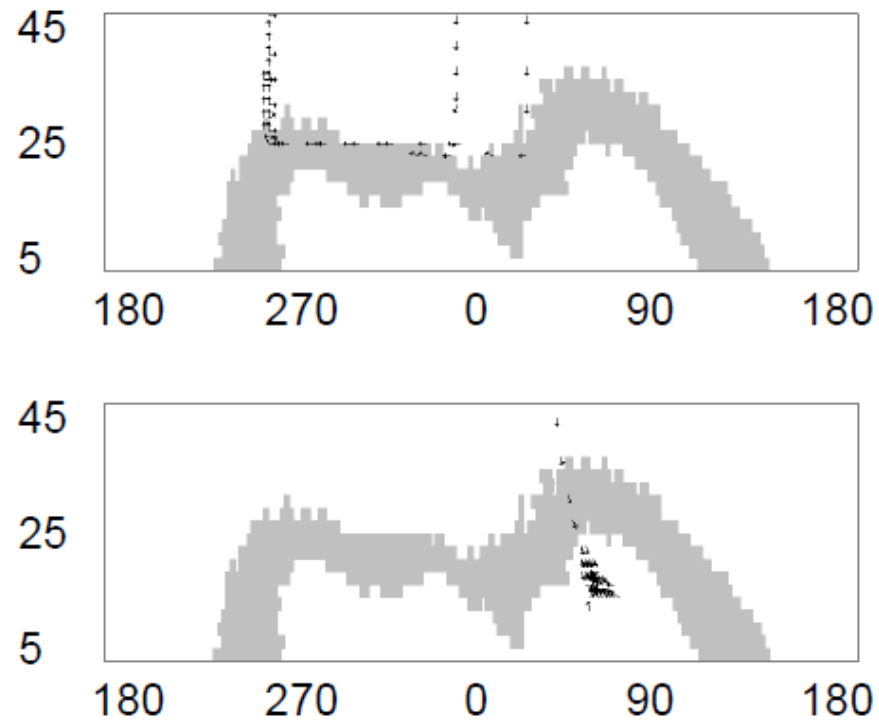
$$GSI(f) = \frac{\sum_{i=1}^N (f(x_i) + f(x'_i) + 1) \bmod 2}{N}$$



GSI over generations



Behavioral classification

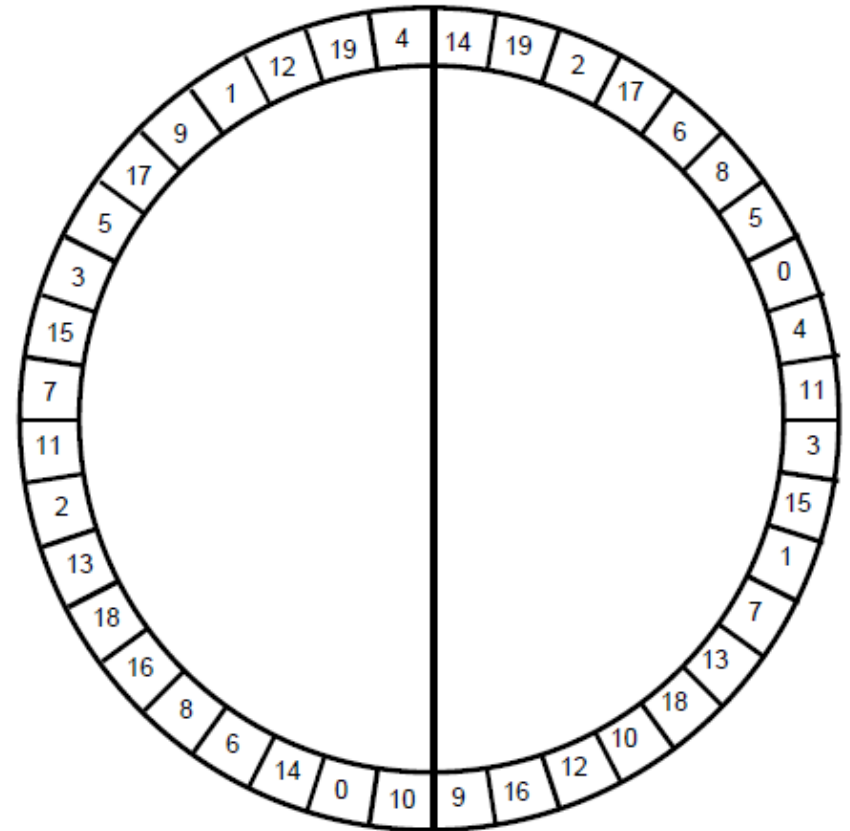


Exercise

- You will have to make a *reactive* action strategy to resolve a task with perceptual ambiguity.
- The reactive strategy is represented by a 2-column table that maps a number (1st column) to a single clockwise or anti-clockwise step (2nd column).
- Max. 10 minutes.

Exercise

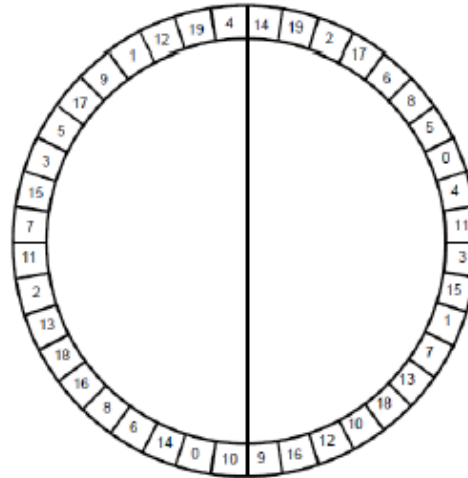
- The “robot” is located in one of the cells.
- It perceives the number in that cell alone.
- Make a reactive strategy that will always bring it to the left half of the circle – given that it always performs 50 steps.



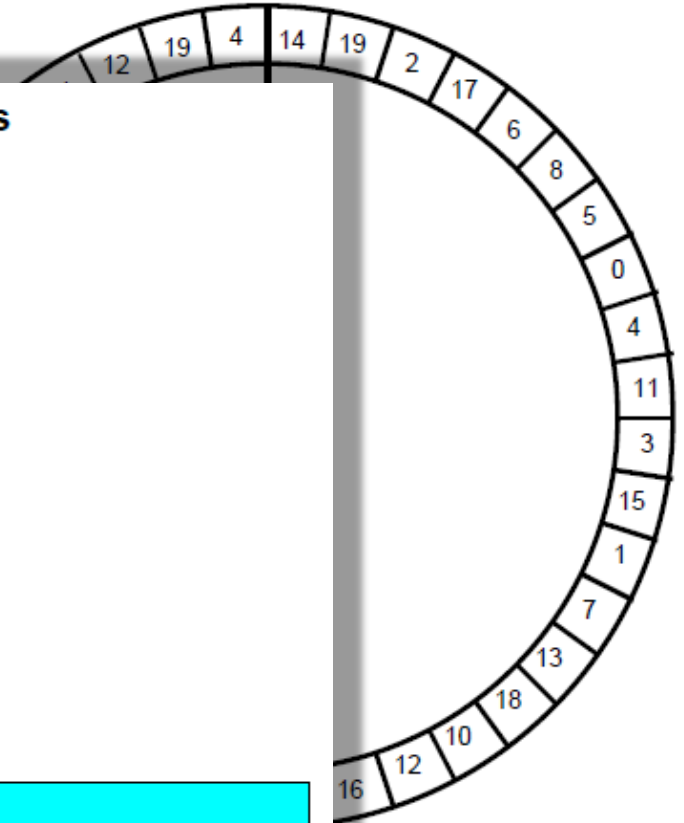
Exercise

- The “robot” is located at the top of the arena.
- It perceives the number of the cell it is currently in.
- Make a reactive strategy that always perceives the robot to the left.

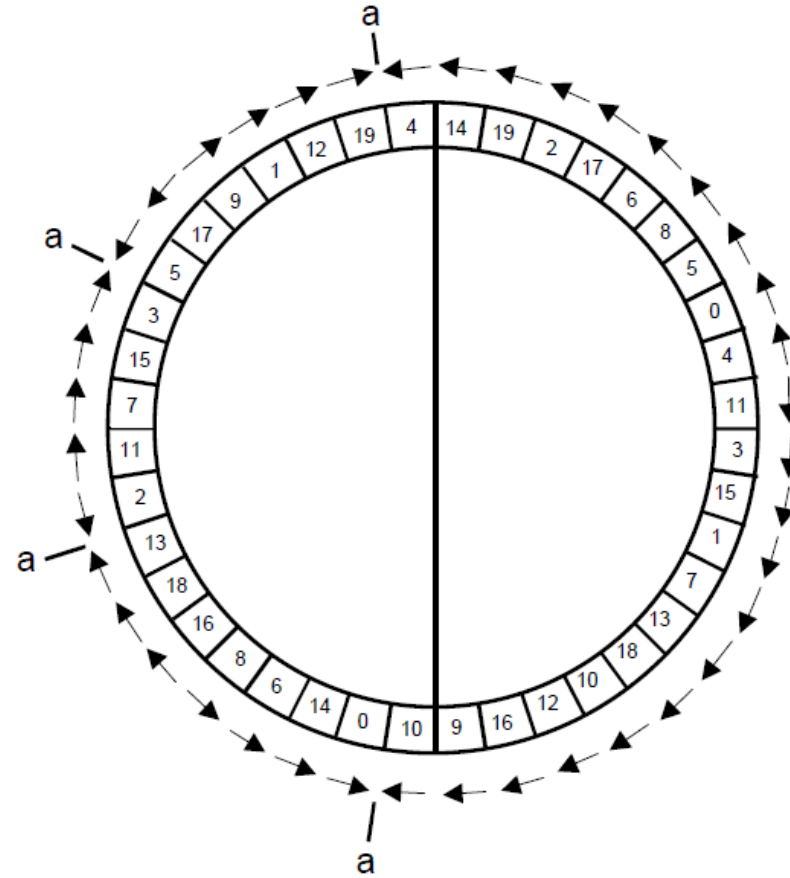
Exercise Evolutionary robotics



State	Action (CW / CCW)
0	
1	
2	
3	
4	
5	



Possible solution – the best?



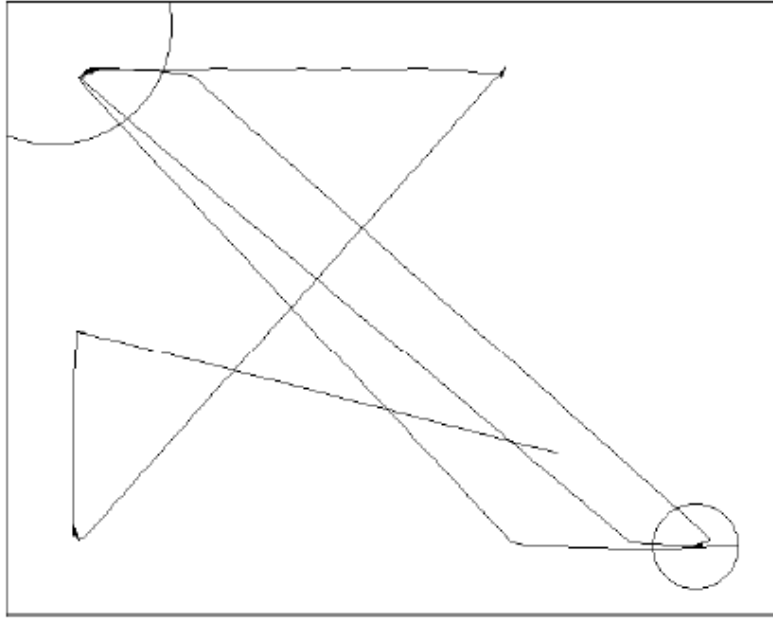
Nolfi, Stefano. "**Power and the limits of reactive agents.**" *Neurocomputing* 42.1 (2002): 119-145.

SMC for classification

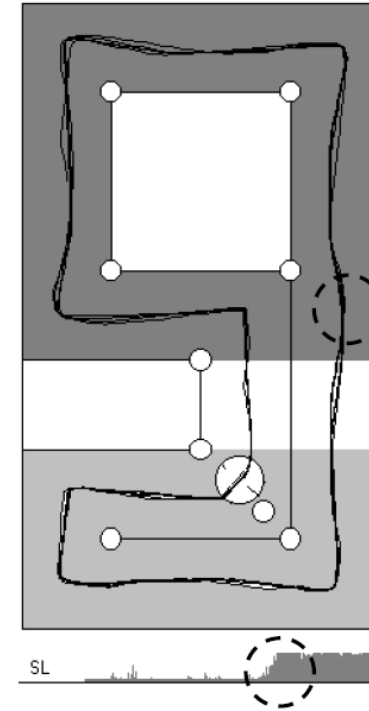
- Use actions to disambiguate sensory inputs
- Behavioral attractors
- Taking the best action *given* the ambiguity

Limitations reactive agents

1.



2.



1. Nolfi, Stefano. **Power and the limits of reactive agents**. Neurocomputing 42, 1 (2002): 119-145.
2. (2004), de Croon G., Nolfi S., Postma E.O., **Toward pro-active embodied agents: On the importance of neural mechanisms suitable to process information in time**. In D. Braha, A. Minai, Y. Bar-Yam (Eds.) Complex Engineering Systems, Perseus Books Groups Press

Pro-active Controllers

Limitations reactive agents

“... limitations of reactive systems are often used to criticize situated approaches more generally.

Strictly speaking, however, internal state is the only requirement for such non-reactive behavior ...”

Randall Beer in “The dynamics of active categorical perception in an evolved model agent.”

Pro-active controllers

Pro-active controllers:

Controllers that have an internal state.

Actions $\vec{a}$ are a function of the sensory inputs $\vec{l}$ and an internal state $\vec{s}$:

$$\begin{aligned}\vec{a} &= f(\vec{l}, \vec{s}) \\ \vec{s} &= g(\vec{l}, \vec{s})\end{aligned}$$



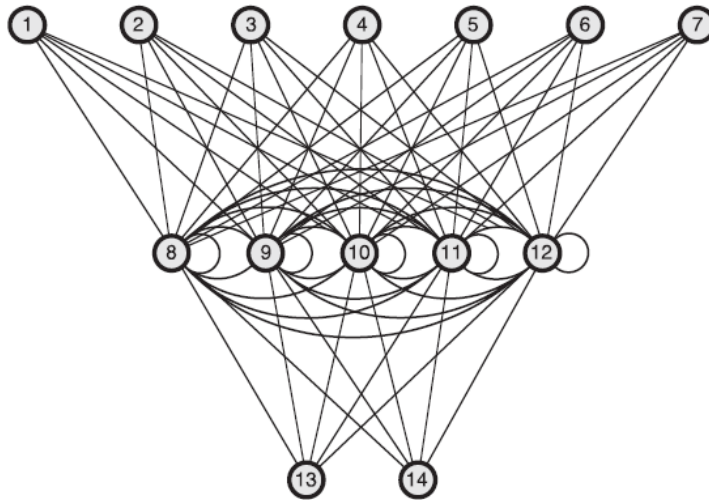
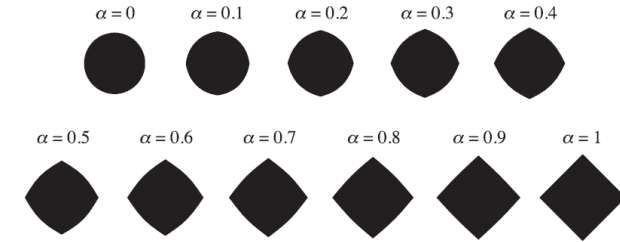
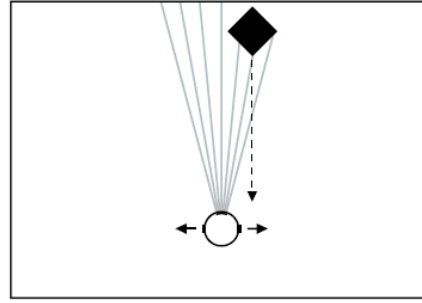
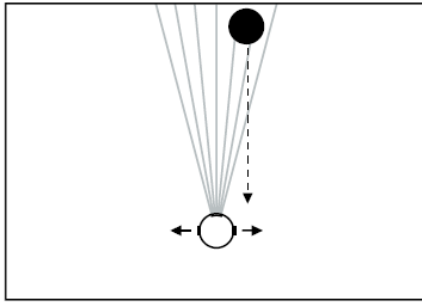
Oranda goldfish

Dynamics of active categorical perception

- Dynamics: the environment and the neural network activations have (coupled) dynamics
- Active: the agent is situated
- Categorical perception: sorting continuous signals received by sensors into discrete categories

Beer, Randall D. **The dynamics of active categorical perception in an evolved model agent.**
Adaptive Behavior 11.4 (2003): 209-243.

Minimally cognitive behavior



CTRNN:

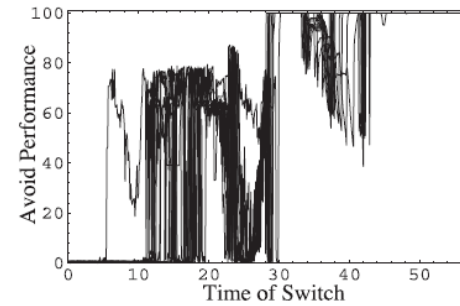
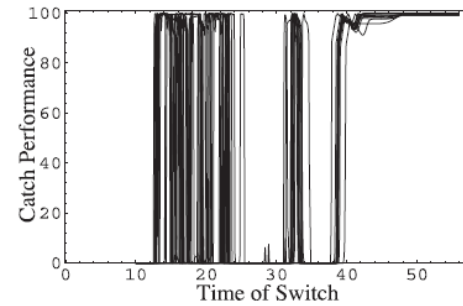
$$\tau_i \dot{s}_i = -s_i + \sum_{j=1}^N w_{ji} \sigma(g_j(s_j + \theta_j)) + I_i$$

$$i = 1, \dots, N$$

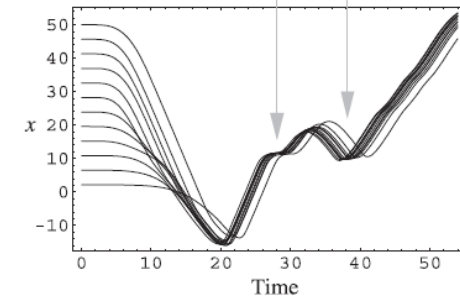
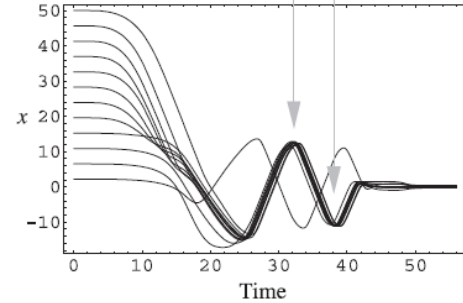
When is the decision made?



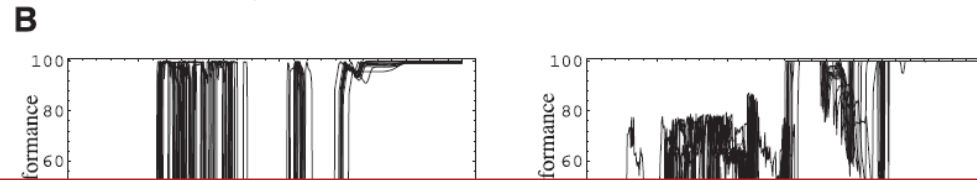
B



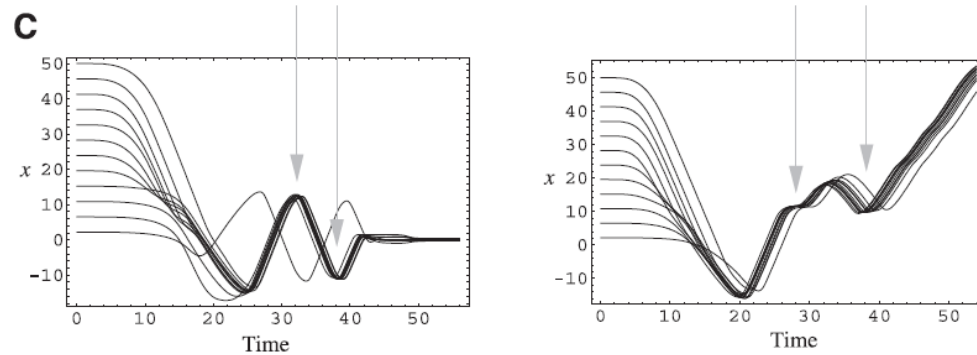
C



When is the decision made?

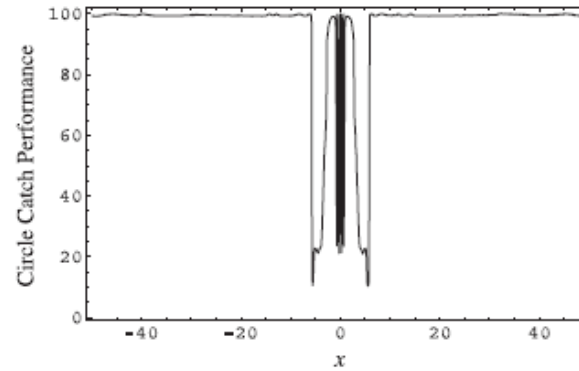


Decisions are temporally extended processes shaped by both sensory signals and neuronal state

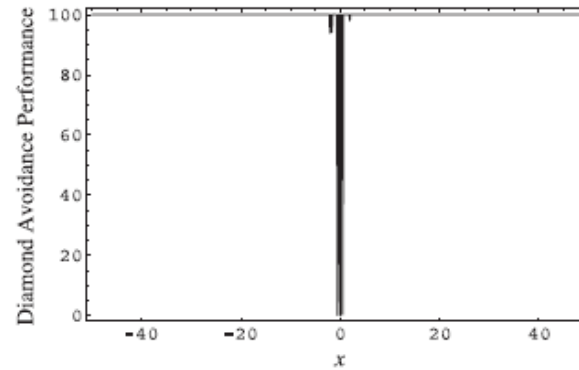


Performance vs. initial location

A

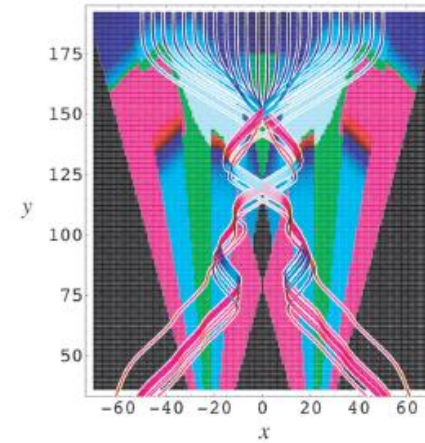
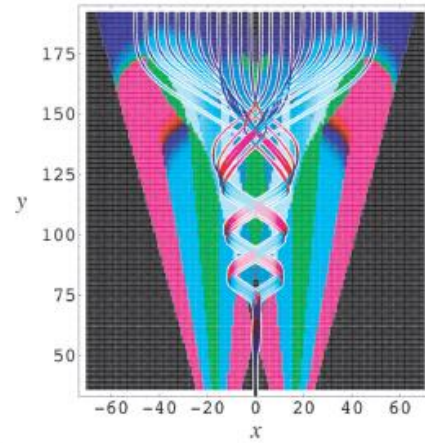


B

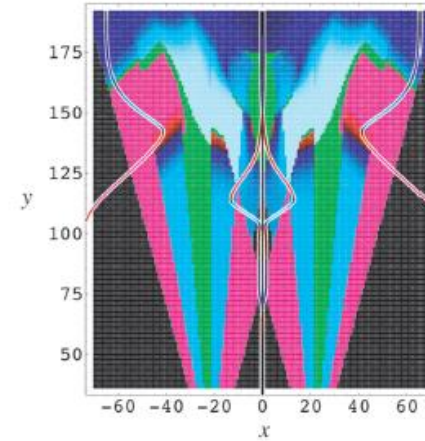
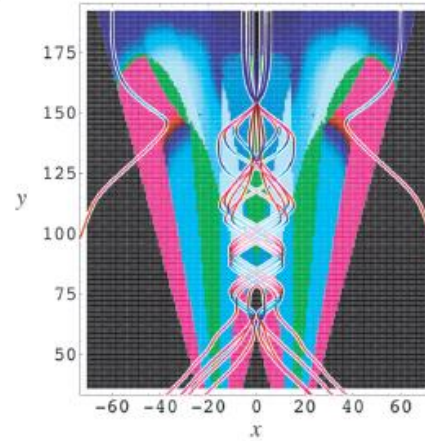


Steady-state analysis

B



C

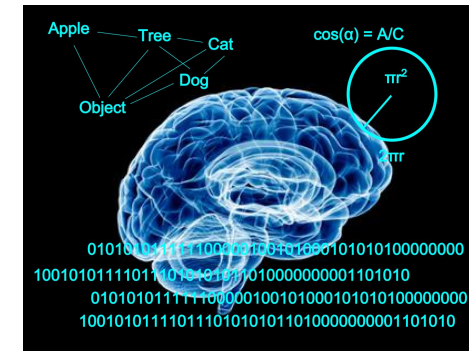


Neural mechanisms

What are the neural mechanisms underlying the behavior of the dynamical system?

Representation?

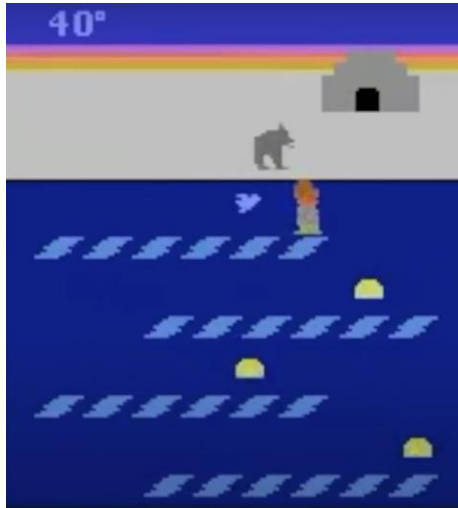
- No neural correlate of a decision, nor of circle / diamond classes.
- Neural state cannot be understood in isolation of world state (x, y).



?

What about deep nets?

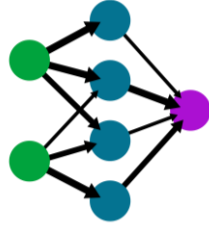
What about deep nets?



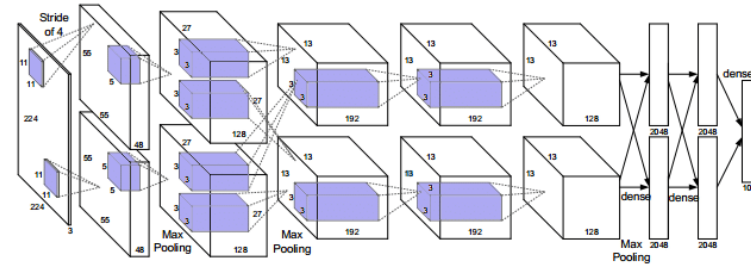
[The GA playing frostbite](#)

	DQN	ES	A3C	RS	GA	GA
Frames	200M	1B	1B	1B	1B	6B
Time	~7-10d	~ 1h	~ 4d	~ 1h or 4h	~ 1h or 4h	~ 6h or 24h
Forward Passes	450M	250M	250M	250M	250M	1.5B
Backward Passes	400M	0	250M	0	0	0
Operations	1.25B U	250M U	1B U	250M U	250M U	1.5B U
amidar	978	112	264	143	263	377
assault	4,280	1,674	5,475	649	714	814
asterix	4,359	1,440	22,140	1,197	1,850	2,255
asteroids	1,365	1,562	4,475	1,307	1,661	2,700
atlantis	279,987	1,267,410	911,091	26,371	76,273	129,167
enduro	729	95	-82	36	60	80
frostbite	797	370	191	1,164	4,536	6,220
gravitar	473	805	304	431	476	764
kangaroo	7,259	11,200	94	1,099	3,790	11,254
seaquest	5,861	1,390	2,355	503	798	850
skiing	-13,062	-15,443	-10,911	-7,679	†-6,502	†-5,541
venture	163	760	23	488	969	†1,422
zaxxon	5,363	6,380	24,622	2,538	6,180	7,864

Size of the network?



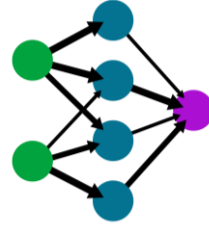
Small ANN



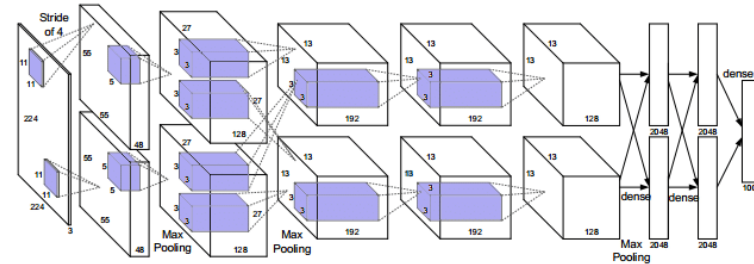
Deep ANN

Characteristics of a small neural network:

Size of the network?



Small ANN

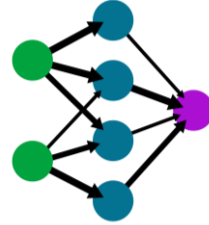


Deep ANN

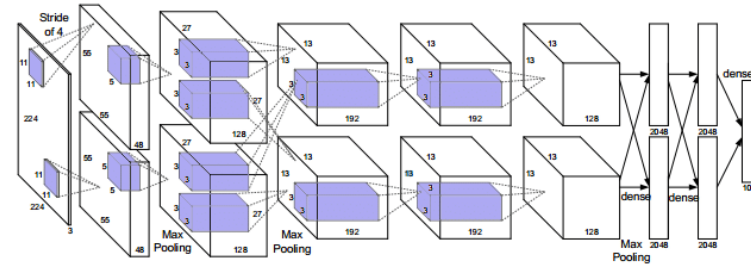
Characteristics of a small neural network:

- **Many fewer calculations**
- **Easier to understand after evolution**

Size of the network?



Small ANN

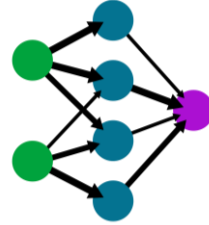


Deep ANN

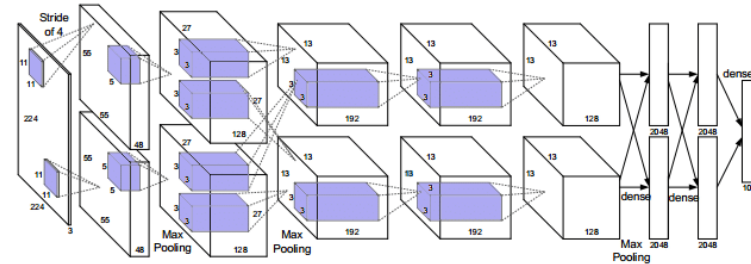
Characteristics of a small neural network:

- **Many fewer calculations**
- **Easier to understand after evolution**
- **Can represent less complex functions, so:**
 - Inputs from high-dimensional sensors have to be pre-processed / abstract
 - Outputs to high-dimensional actuators have to be abstract

Size of the network?



Small ANN



Deep ANN

Characteristics of a small neural network:

- Many fewer calculations
- Easier to train
- Can be deployed on low-power devices

A disadvantage?

- Inputs from high-dimensional sensors have to be pre-processed / abstract
- Outputs to high-dimensional actuators have to be abstract

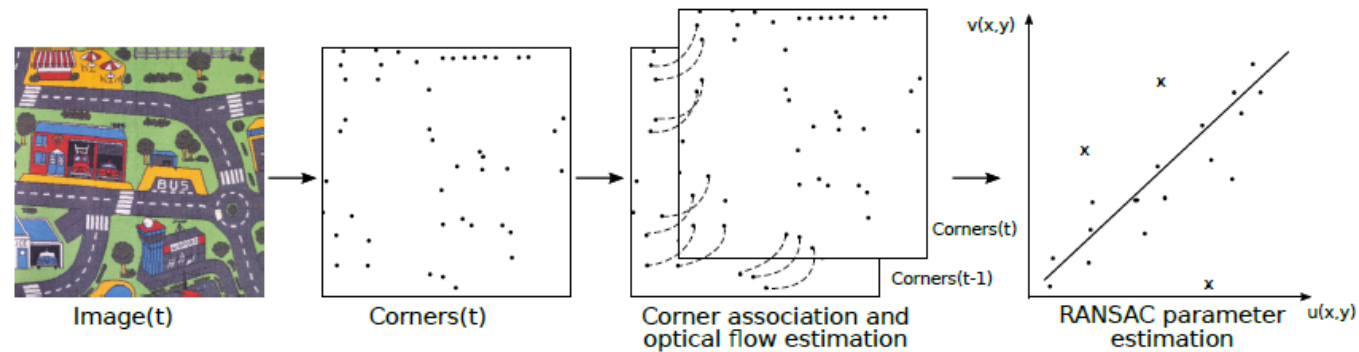
Abstraction as a mechanism to cross the reality gap

Divergence and its time derivative as input

Desired vertical acceleration as output



From images over time to divergence:



Abstraction as a mechanism to cross the reality gap

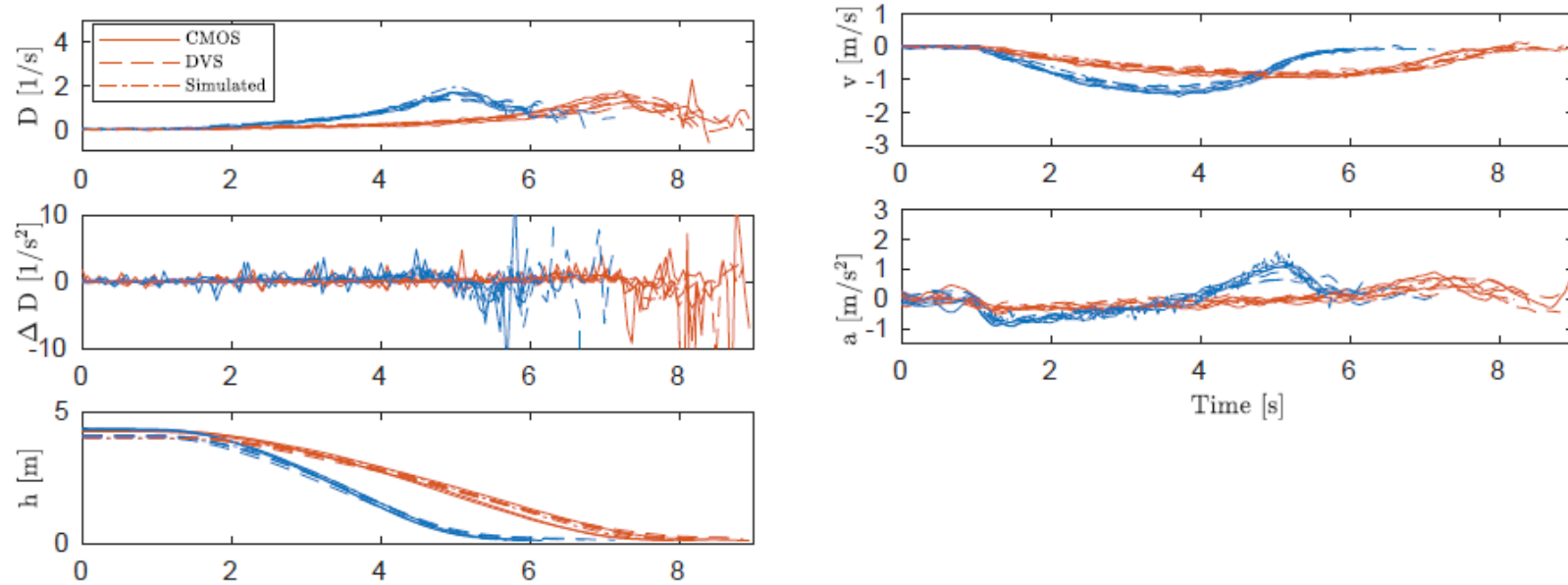
Vary factors during evolution that can be different between simulation and reality:

Table 1

Evolutionary parameters

Name	Value
Number of Generations	250
Number of Runs	4
Range of delays	[1, 4] samples
Range of computational jitter probability	[0, 0.2]
Range of divergence noise (σ_w^2)	[0.05, 0.15] 1/s
Range of divergence noise (σ_p^2)	[0, 0.25] 1/s
Range of thrust time constant (τT)	[0.005, 0.04]
Range of simulation frequency	[30, 50] Hz

Abstraction as a mechanism to cross the reality gap



Evolutionary robotics



Advantages:

- Meta-heuristic optimization allows for inclusion of any kind of parameter (body / sensor processing / ...) and any type of fitness function.
- No explicit assumption on the existence of a state / Markovian property of the state / etc.
- Computational efficiency is ensured by choosing the controller in advance.
- If a solution is found, it is often very interesting – exploitation of sensory-motor coordination.

Evolutionary robotics



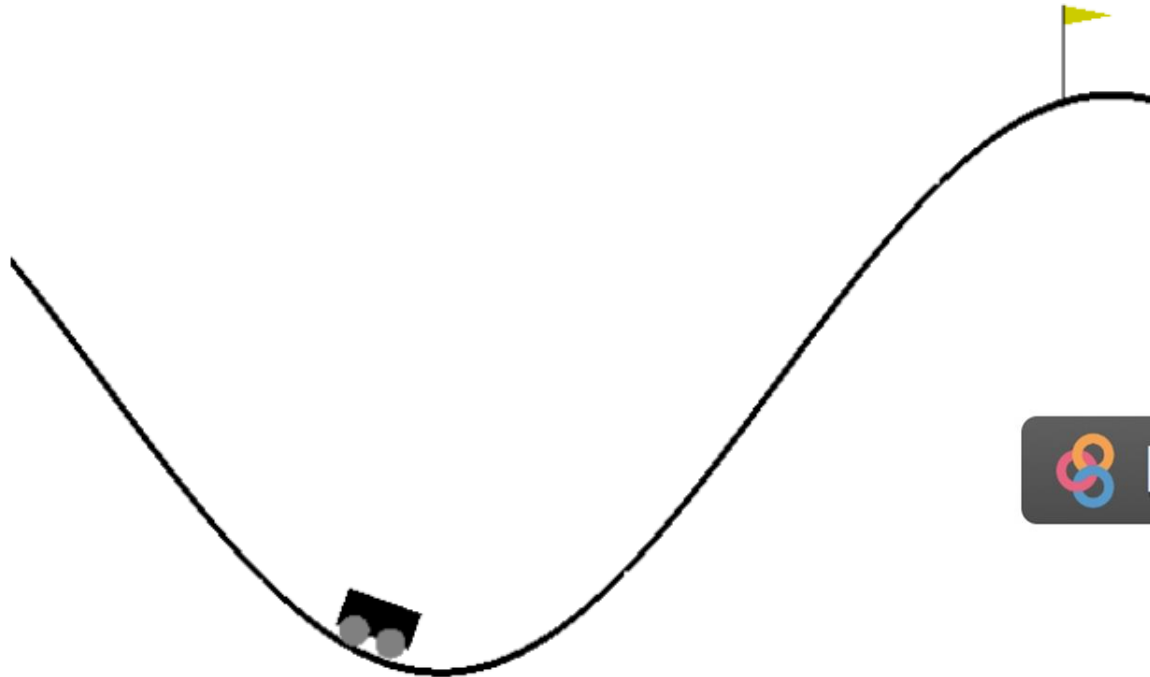
Drawbacks:

- The chosen structure poses implicit constraints on the solution. What are the limitations of a 5-hidden neuron feedforward network?
- If no solution is found, there is often no clearly identifiable reason for this failure.
- Typically no online adaptation (= during operation), how to deal with the reality gap?
- Until now a limited task complexity (not far beyond phototaxis).

Learn to use an EA

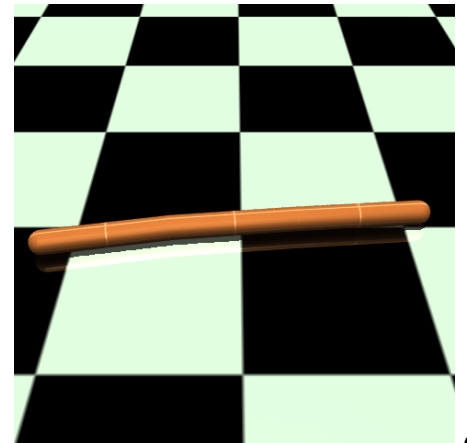
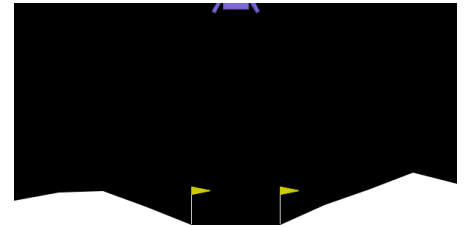
Jupyter Notebook

Use an evolutionary robotics approach for the continuous mountain car problem from [Al gym](#)



Assignment – ideas

- AI gym control problems / games:
 - Super Mario
 - Lunar lander
 - ATARI games
 - Robotic tasks (Mujoco)
- Combination with insect intelligence / swarming / spiking neural nets



Questions?



More information
mavlab.tudelft.nl